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RV COLLEGE OF ENGINEERING®

(An Autonomous Institution Affiliated to VTU)

II Semester B. E. Regular / Supplementary Examinations Aug-2024

(Common to AS, CH, IM, ME)

VECTOR CALCULUS AND COMPUTATIONAL METHODS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

		M	BT	CO
1.1	Normal to the paraboloid of revolution $x^2+y^2=az$ at the point $(a,-a,2a)$ is _____.	02	2	3
1.2	A person on a hang glider is spiraling upward due to rapidly rising air on a path having position vector $r(t)=3\cos t\,\hat{\imath}+3\sin t\,\hat{\jmath}+t^2\hat{k}$. Determine the velocity at $t=\pi/2$.	02	2	2
1.3	If R is the projection of the surface $y=4$ on to the $xz-plane$, then $ds=$ _____.	02	2	1
1.4	If C is the curve $y=x^2$ from $x=0$ to $x=2$ in the $xy-plane$, the $\int_c [x\,dx+y\,dy]=$ _____.	02	3	2
1.5	The partial differential equation obtained by eliminating the arbitrary function form the equation $z=f(x^2-y^2)$ is _____.	02	2	2
1.6	Find one of the solution of the partial differential equation $xp+yq=z$.	02	1	1
1.7	The Taylor series solution of $y'=x^2-y$, $y(1)=1$ up to second degree term is _____.	02	2	2
1.8	Evaluate $\Delta^5[(2-x^2)(ax-3)(bx^2-1)]$, by taking $h=1$.	02	1	1
1.9	The velocity (m/s) of an object at discrete time (sec) is given below: t: 0, 3, 6 v: 1, 10, 15 The distance travelled in 6 sec is _____.	02	2	3
1.10	In Newton-Raphson method, to find the real root of an equation, the curve $y=f(x)$ is replaced by _____.	01	2	4
1.11	Root of the equation $e^x-3x=2$ lies in the finite interval _____.	01	1	1

PART-B

			M	BT	CO
2	a	Find the directional derivative of $f=xy+yz+zx$ at the point $(1,2,0)$ in the direction of tangent to the curve $\vec{r}(t)=2t^2\hat{\imath}+(t^2-4t)\hat{\jmath}+(3t-5)\hat{k}$ at $t=1$.	08	3	3
	b	Find the value of m such that $\vec{F}=(mxy-z^3)\hat{\imath}+(m-2)x^2\hat{\jmath}+(1-m)xz^2\hat{k}$ is irrotational and determine scalar potential ϕ such that $\nabla\phi=\vec{F}$.	08	3	2
3	a	Find the work done in moving a particle in the force field $\vec{F}=(2y+3)\hat{\imath}+xz\hat{\jmath}+(yz-x)\hat{k}$ along the curve $C: x=2t^2$, $y=t$, $z=t^3$ from $(0,0,0)$ to $(2,-1,-1)$.	06	3	2

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	b	Verify Stokes' theorem for $\vec{F} = z\hat{\imath} + x\hat{\jmath} + 2y\hat{k}$ and for the rectangle in the plane $x = y$ with vertices $(0,0,0), (1,1,0), (1,1,2), (0,0,2)$	10	3	3
		OR			
4	a	Using Green's theorem in the plane, evaluate $\oint_c [(xy + y^2)dx + x^2dy]$ taken around the boundary region enclosed between the curves $y = x$ and $4y = x^2$	08	3	2
	b	Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ over the surface bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$ using Gauss divergence theorem.	08	4	4
5	a	Form a partial differential equation which represent the family of sphere with center on $XY - plane$ and having a fixed radius r.	04	2	2
	b	Solve: $(y^2 + z^2)p + x(yq - z) = 0$.	06	3	3
	c	Solve by the method of separation of variables $u_{xx} - 2u_x + u_y = 0$.	06	3	2
		OR			
6	a	Solve the Lagrange linear partial differential equation: $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$.	06	3	3
	b	Obtain the various possible solutions of one dimensional heat equation in by the method of separation of variables and discuss about the suitable solution.	10	4	3
7	a	Using Newton-Raphson method, find a positive real root of the equation $e^x - \sin x = 5$ correct to four places of decimals with 0.5 as initial approximation for the solution.	05	3	3
	b	Compute $y(1.2)$ by Taylor's series method if $y' = x^2y - 1$ and $y(1) = 1$, by considering up to third order term.	05	2	2
	c	Apply Milne's method to solve the differential equation, $\frac{dy}{dx} = y - \frac{2x}{y}$, for $x = 0.8$. Given that,	06	3	3

x	0	0.2	0.4	0.6
y	1	1.1832	1.3416	1.4832

		OR			
8	a	Using false position method find an approximate real root of $x^3 + x - 1 = 0$. Perform three iterations.	06	2	3
	b	Using fourth order Runge-Kutta method find y at $x = 1.2$ taking step size as 0.1. Given that, $\frac{dy}{dx} = -2xy^2$, $y(1) = 1$.	10	3	4
9	a	The following data is on melting point of an alloy of lead and zinc, where t is temperaturein Celsius and p is the percentage of lead in the alloy.			

p	40	50	60	70	80	90
t	180	204	226	250	276	304

		Find the melting point of an alloy containing 45% and 86% of lead using suitable interpolation formula.	08	2	2
	b	Determine a polynomial $y(x)$ passing through the points $(2,1)$, $(4,7)$, and $(7,15)$ using suitable interpolation technique and hence find $y(4.5)$.	08	2	3
		OR			

10	a	Evaluate $\int_0^1 \frac{dx}{4x+5}$ by dividing the interval into six equal subintervals using: i) Simpson's $1/3^{rd}$ rule ii) Simpson's $3/8^{th}$ rule, and iii) Weddle's rule	08	2	1
	b	Construct finite difference table of the function $f(x) = \sqrt{x}$ for $x = 15, 17, 19, 21, 23, 25$ (use three decimal places). Using suitable numerical differentiation technique determine $f'(15)$ and $f''(23)$.	08	2	1

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VECTOR CALCULUS AND COMPUTATIONAL METHODS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.
3. Use of scientific calculator is permitted.
4. Use of mathematics Handbook is permitted. Do not write anything on handbook.

PART-A

1	1.1	A particle moves along the curve $x = e^{-t}, y = 2\cos 3t, z = 2\sin 3t$, the velocity vector at $t = 0$ is _____.	02
	1.2	The unit normal vector to the surface $3x^2 + y^2 + z^2 = 0$ at $(-1,-1,2)$ is _____.	02
	1.3	If $\vec{F} = y^2\hat{\imath} + z^2\hat{\jmath} + x^2\hat{k}$ and V is the region bounded by the cube $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$, then $\iiint_V \nabla.\vec{F}\, dv =$ _____.	02
	1.4	The value of $\int_C x\,dy - y\,dx$ around the circle C whose equation is $x^2 + y^2 = 1$ is _____.	02
	1.5	The partial differential equation obtained from $z = (x + a)(y + b)$ by eliminating a and b is _____.	02
	1.6	The order and degree of partial differential equation $\frac{\partial z}{\partial y} + 2\left(\frac{\partial^2 z}{\partial x^2}\right)^2 - \left(\frac{\partial^3 z}{\partial x^3}\right)^2 + z = 1$ is _____ and _____ respectively.	02
	1.7	The first approximate positive root of $x\sin x + \cos x = 0$ by taking $x_0 = \pi$ by Newton-Raphson method is _____.	02
	1.8	The first approximate to a real root of the equation $2x - \log_{10} x - 7 = 0$ correct to 3 decimal places by the method of false position between 3.5 and 4 is _____.	02
	1.9	Construct the difference table for the following data: <table><tr><td>x</td><td>1</td><td>2</td><td>4</td><td>6</td></tr><tr><td>y</td><td>5</td><td>3</td><td>4</td><td>2</td></tr></table>	02
	1.10	The value of $\Delta^4[(3 + x)(1 + 4x^2)(5 - 2x)]$ taking the interval of differencing $h = 1$ is _____.	02

PART-B

2	a	Obtain the directional derivative of $f(x,y,z) = xy^2 - 4x^2y + z^2$ at $(1,-1,2)$ in the direction $6\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$.	08
	b	Show that $\vec{F} = (e^x\cos y + yz)\hat{\imath} + (xz - e^x\sin y)\hat{\jmath} + (xy + z)\hat{k}$ is conservative and find its scalar potential.	08
3	a	If $\vec{F} = 18z\hat{\imath} - 12\hat{\jmath} + 3y\hat{k}$ and S is the part of the plane $2x + 3y + 6z = 12$ in the first octant, then find $\iint_S \vec{F}.\hat{n}\, dS$.	06
	b	Verify Green's theorem in the plane for $\int_C (e^x\sin y)dx + (e^{-x}\cos y)dy$ where C is the rectangle whose vertices are $(0,0), (\pi,0), \left(\pi,\frac{\pi}{2}\right), \left(0,\frac{\pi}{2}\right)$	10

		OR	
4	a	Verify Stoke's theorem for the vector filed $\vec{F} = (x^2 - y^2)\hat{\imath} + 2xy\,\hat{\jmath}$ in the rectangular region bounded by the lines $x = 0, x = 2, y = 0, y = 4$.	08
	b	Evaluate $\int_C \vec{F}.d\vec{r}$ over the circular path $x^2 + y^2 = a^2$ where $\vec{F} = \sin y\ \hat{\imath} + x(1 + \cos y)\,\hat{\jmath}$.	08
5	a	Form a *PDE* by eliminating the arbitrary functions ϕ and Ψ from the relation $z = x\,\phi(y) + y\,\Psi(x)$.	06
	b	Solve $4u_x + u_y = 3u$ by the method of separation of variables, given that $u(0, y) = 2e^{5y}$.	10
		OR	
6	a	Solve $(y^2 + z^2)p + xyq = xz$	06
	b	Obtain the various possible solutions of heat equation in one dimension by the method of separation of variables.	10
7	a	Find a real root of the equation $xe^x - 2 = 0$ using method of false position method correct to 3 decimal places. Carry out 3 iterations.	08
	b	Solve by Taylor series method the value of y at $x = 0.1$ correct to 3 decimal places if $y(x)$ satisfies $\frac{dy}{dx} = x^2y - 1$ and $y(0) = 1$.	08
		OR	
8	a	Obtain the root of the equation $\sin x = 1 + x^3$ using Newton-Raphson method correct to 4 decimal places. Perform 4 iterations.	08
	b	Solve the differential equation $\frac{dy}{dx} = \frac{y-x}{y+x}$ for $x = 0.2$ using Runge-Kutta 4th order method given $x_0 = 0, y_0 = 1$.	08
9	a	Details regarding marks scored by 215 candidates in an examination are given by the following table. Using Newton-Gregory interpolation formula, estimate the number of candidates who secured marks not more than 45.	08

Marks	30 – 40	40 – 50	50 – 60	60 – 70	70 – 80
Number of students	35	48	70	40	22

	b	Evaluate $\int_0^{0.6} e^{-x^2}\,dx$ using i) Simpson's 1/3 rule ii) Simpson's 3/8 rule iii) Weddle's rule Taking 6 sub-intervals.	08
		OR	
10	a	A rod is rotating in a plane. The following table gives the angle θ in radians through which the rod has turned for various values of the time *t seconds*.	

t	0	0.2	0.4	0.6	0.8	1	1.2
θ	0	0.12	0.49	1.12	2.02	3.20	4.67

		Calculate the angular velocity and angular acceleration of the rod when $t = 0.4$ *seconds*.	08
	b	Using suitable interpolation formula, fit a polynomial for the data	

x	0	1	3	4
y	-12	0	6	12

		Hence estimate y at $x = 2$.	08

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PART-A

			M	BT	CO
1	1.1	Normal to the paraboloid of revolution $x^2+y^2=az$ at the point $(a,-a,2a)$ is _____.	02	2	3
	1.2	A person on a hang glider is spiraling upward due to rapidly rising air on a path having position vector $r(t)=3\cos t\,\hat{\imath}+3\sin t\,\hat{\jmath}+t^2\hat{k}$. Determine the velocity at $t=\pi/2$.	02	2	2
	1.3	If R is the projection of the surface $y=4$ on to the $xz-plane$, then $ds=$ _____.	02	2	1
	1.4	If C is the curve $y=x^2$ from $x=0$ to $x=2$ in the $xy-plane$, the $\int_c [x\,dx+y\,dy]=$ _____.	02	3	2
	1.5	The partial differential equation obtained by eliminating the arbitrary function form the equation $z=f(x^2-y^2)$ is _____.	02	2	2
	1.6	Find one of the solution of the partial differential equation $xp+yq=z$.	02	1	1
	1.7	The Taylor series solution of $y'=x^2-y$, $y(1)=1$ up to second degree term is _____.	02	2	2
	1.8	Evaluate $\Delta^5[(2-x^2)(ax-3)(bx^2-1)]$, by taking $h=1$.	02	1	1
	1.9	The velocity (m/s) of an object at discrete time (sec) is given below: t: 0, 3, 6 v: 1, 10, 15 The distance travelled in 6 sec is _____.	02	2	3
	1.10	In Newton-Raphson method, to find the real root of an equation, the curve $y=f(x)$ is replaced by _____.	01	2	4
	1.11	Root of the equation $e^x-3x=2$ lies in the finite interval _____.	01	1	1

PART-B

2	a	Find the directional derivative of $f=xy+yz+zx$ at the point $(1,2,0)$ in the direction of tangent to the curve $\vec{r}(t)=2t^2\hat{\imath}+(t^2-4t)\hat{\jmath}+(3t-5)\hat{k}$ at $t=1$.	08	3	3
	b	Find the value of m such that $\vec{F}=(mxy-z^3)\hat{\imath}+(m-2)x^2\hat{\jmath}+(1-m)xz^2\hat{k}$ is irrotational and determine scalar potential ϕ such that $\nabla\phi=\vec{F}$.	08	3	2
3	a	Find the work done in moving a particle in the force field $\vec{F}=(2y+3)\hat{\imath}+xz\hat{\jmath}+(yz-x)\hat{k}$ along the curve $C: x=2t^2$, $y=t$, $z=t^3$ from $(0,0,0)$ to $(2,-1,-1)$.	06	3	2

	b	Verify Stokes' theorem for $\vec{F} = z\hat{\imath} + x\hat{\jmath} + 2y\hat{k}$ and for the rectangle in the plane $x = y$ with vertices $(0,0,0), (1,1,0), (1,1,2), (0,0,2)$	10	3	3
		OR			
4	a	Using Green's theorem in the plane, evaluate $\oint_c [(xy + y^2)dx + x^2dy]$ taken around the boundary region enclosed between the curves $y = x$ and $4y = x^2$	08	3	2
	b	Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ over the surface bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $2x + 2y + z = 4$ using Gauss divergence theorem.	08	4	4
5	a	Form a partial differential equation which represent the family of sphere with center on $XY - plane$ and having a fixed radius r.	04	2	2
	b	Solve: $(y^2 + z^2)p + x(yq - z) = 0$.	06	3	3
	c	Solve by the method of separation of variables $u_{xx} - 2u_x + u_y = 0$.	06	3	2
		OR			
6	a	Solve the Lagrange linear partial differential equation: $x^2(y - z)p + y^2(z - x)q = z^2(x - y)$.	06	3	3
	b	Obtain the various possible solutions of one dimensional heat equation in by the method of separation of variables and discuss about the suitable solution.	10	4	3
7	a	Using Newton-Raphson method, find a positive real root of the equation $e^x - \sin x = 5$ correct to four places of decimals with 0.5 as initial approximation for the solution.	05	3	3
	b	Compute $y(1.2)$ by Taylor's series method if $y' = x^2y - 1$ and $y(1) = 1$, by considering up to third order term.	05	2	2
	c	Apply Milne's method to solve the differential equation, $\frac{dy}{dx} = y - \frac{2x}{y}$, for $x = 0.8$. Given that,	06	3	3
		OR			
8	a	Using false position method find an approximate real root of $x^3 + x - 1 = 0$. Perform three iterations.	06	2	3
	b	Using fourth order Runge-Kutta method find y at $x = 1.2$ taking step size as 0.1. Given that, $\frac{dy}{dx} = -2xy^2$, $y(1) = 1$.	10	3	4
9	a	The following data is on melting point of an alloy of lead and zinc, where t is temperaturein Celsius and p is the percentage of lead in the alloy.			
	b	Find the melting point of an alloy containing 45% and 86% of lead using suitable interpolation formula.	08	2	2
		Determine a polynomial $y(x)$ passing through the points $(2,1)$, $(4,7)$, and $(7,15)$ using suitable interpolation technique and hence find $y(4.5)$.	08	2	3
		OR			

x	0	0.2	0.4	0.6
y	1	1.1832	1.3416	1.4832

p	40	50	60	70	80	90
t	180	204	226	250	276	304

10	a	Evaluate $\int_0^1 \frac{dx}{4x+5}$ by dividing the interval into six equal subintervals using: i) Simpson's $1/3^{rd}$ rule ii) Simpson's $3/8^{th}$ rule, and iii) Weddle's rule	08	2	1
	b	Construct finite difference table of the function $f(x) = \sqrt{x}$ for $x = 15, 17, 19, 21, 23, 25$ (use three decimal places). Using suitable numerical differentiation technique determine $f'(15)$ and $f''(23)$.	08	2	1